The Open Group Preliminary Standard

O-PAS™ Standard, Version 2.1

Part 6.3 – Information and Exchange Models: Alarm and Events Configuration

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The Open Group Preliminary Standard

O-PAS™ Standard, Version 2.1: Part 6.3 – Information and Exchange Models: Alarm and Events Configuration

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Contents

1 Introduction 1
- 1.1 Objective 1
- 1.2 Overview 1
- 1.3 Conformance 1
- 1.4 Normative References 1
- 1.5 Terminology 2
- 1.6 Future Directions (Informative) 3

2 Terms and Definitions 4
- 2.1 OPAF Glossary and Abbreviations 4

3 Information Model 5
- 3.1 Simplified Data Flow Information Model 5
- 3.2 Alarm System and Related State Models 5
- 3.3 Alarm Configuration 6
  - 3.3.1 Overview 6
  - 3.3.2 Alarm Attributes Common to all Alarm Types 6
  - 3.3.3 Alarm Attributes for Specific Alarm Types 7
  - 3.3.4 Certification Expiration Alarm Configuration 8

4 Import/Export Exchange Format 9
- 4.1 Overview 9
- 4.2 AML Specifications 9
  - 4.2.1 Overview 9
  - 4.2.2 AML Interfaces Used for Alarm Configuration 12
  - 4.2.3 AML Attributes for Condition Classes 13
  - 4.2.4 Alarms in the AML InstanceHierarchy 14
- 4.3 Profiles and Conformance Units 15

5 Download/Upload Exchange Models 18

6 OCF Information Model 19
- 6.1 Overview 19

7 Namespaces 20
- 7.1 Namespace Metadata 20
- 7.2 Handling of OPC UA Namespaces 20

A Alarm Configuration Details and Approach (Informative) 23
- A.1 Alarm Types 23
- A.2 Mapping FF Alarm Types to OPC UA Alarm Types 24
- A.3 Standard Alarm Fields 25
- A.4 Identifying the Configurable Parameters for the OPC UA Alarms 25

B AML Definitions (Normative) 34

C Abbreviations 35

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Preface

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This Document

This document is Part 6.3 of the O-PAS™ Standard, Version 2.1, a Preliminary Standard of The Open Group. It has been developed and approved by The Open Group.

The O-PAS Standard consists of the following 12 parts (of the anticipated 13 parts):

- O-PAS Part 1 – Technical Architecture Overview (Informative)
- O-PAS Part 2 – Security
- O-PAS Part 3 – Profiles (Informative)
- O-PAS Part 4 – O-PAS Connectivity Framework (OCF)
- O-PAS Part 5 – System Management
- O-PAS Part 6.1 – Information and Exchange Models: Overview and Interfaces

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- O-PAS Part 6.2 – Information and Exchange Models: Basic Configuration
- O-PAS Part 6.3 – Information and Exchange Models: Alarm and Events Configuration (this document)
- O-PAS Part 6.4 – Information and Exchange Models: Function Blocks
- O-PAS Part 6.5 – Information and Exchange Models: IEC 61499
- O-PAS Part 6.6 – Information and Exchange Models: IEC 61131
- O-PAS Part 7 – Distributed Control Platform

The parts listed above are separate documents that can be updated and re-versioned as required as we move forward with the O-PAS Standard.

The O-PAS Standard, Version 2.1 is being published initially as a Preliminary Standard since it addresses an emerging area of technology; therefore, it may change before being published as a full standard of The Open Group. In such a case it will be made as upwards-compatible as possible with the corresponding Preliminary Standard, but complete upwards-compatibility is not guaranteed.

Once this document becomes a full standard, it will supersede the O-PAS Standard, Version 1.0 published in December 2019.

Conventions

A Glossary and Abbreviations reference is available. If a term is not defined in that document then the common English definition, as defined in Merriam-Webster's Collegiate Dictionary, applies.

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The Open Group gratefully acknowledges the International Society of Automation for use of the ISA copyrighted ANSI/ISA-18.2 and ANSI/ISA 62443 series of standards. Visit www.isa.org.

The Open Group gratefully acknowledges and thanks the OPC Foundation for the template used to develop an OPC UA Companion Specification that is included in this O-PAS Standard. Visit www.opcfoundation.org.

The Open Group gratefully acknowledges the contribution of the following people in the development of this document:

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The Open Group gratefully acknowledges the contribution and leadership of the Co-Chairs of the Open Process Automation™ Forum and Working Groups.

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Referenced Documents

The following documents are referenced in Part 6.3 of the O-PAS Standard.

The documents are referred to in the text in such a way that some or all of their content constitutes requirements of this part. For dated references, only the version cited applies. For undated references, the latest version of the referenced document (including any amendments) applies.

(Please note that the links below are good at the time of writing but cannot be guaranteed for the future.)

Normative References

Normative references for Part 6.3 of the O-PAS Standard are defined in Section 1.4.

Informative References

- ANSI/ISA-18.2:2016: Management of Alarm Systems for the Process Industries, Alarm Systems – A Guide to Design, Management, and Procurement, Engineering Equipment Materials Users' Association (EEMUA), Publication No. 191, 3rd Edition, London, UK (2013)
- ANSI/ISA-TR62443-2-3:2015: Security for Industrial Automation and Control Systems – Part 2-3: Patch Management in the IACS Environment (adopted by IEC as IEC TR 62443-2-3)
- Extensible Markup Language (XML) 1.0 (Third Edition), W3C Recommendation 04 February 2004, available at: http://www.w3.org/TR/2004/REC-xml-20040204/
- FieldComm Group FF-890: FOUNDATION™ Fieldbus Specification – Function Block Application Process Part 1 (August 2012); refer to: https://fieldcomm-group.myshopify.com/collections/types/products/ff-spec

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1 Introduction

1.1 Objective

The objective of Part 6.3 of the O-PAS Standard is to define the import/export information model used to configure alarm functions in O-PAS components. It defines the download/upload information model used to download and upload the alarm configuration information to O-PAS components, and it defines the runtime information model used to exchange and manage alarms among O-PAS components.

1.2 Overview

Part 6.3 of the O-PAS Standard defines the use of the OPC UA Part 9 information models to support configuration management of alarms and events within a system made up of O-PAS conformant components. This configuration supports the parameters for the management of alarms and events in an O-PAS system. This document is restricted to a limited subset of possible alarm types. Additional alarming and event functionality will be added in a future version.

This document covers alarm configuration and the related state model of alarms and events. The OPC UA Part 9 implementation of the data model is used in this document.

The focus of Part 6.3 is on the OPC UA server side of the information model. Requirements related to the client side (HMI components and reporting) are not covered.

1.3 Conformance

Part 6.3 of the O-PAS Standard is normative, and conformance requirements (marked with "CRQ" identifiers) are defined later in Chapters 3, 4, 6, and 7.

1.4 Normative References

The following standards contain provisions which, through references in this document, constitute provisions of Part 6.3 of the O-PAS Standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the standards listed below.

- IEC 62714-1:2018: Engineering Data Exchange Format for Use in Industrial Automation Systems Engineering – Automation Markup Language (AutomationML) – Part 1: Architecture and General Requirements; refer to: https://webstore.iec.ch/publication/32339

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- IETF RFC 2119: Key Words for Use in RFCs to Indicate Requirement Levels, March 1997; refer to: https://tools.ietf.org/html/rfc2119
- ISO/IEC 29500-2: Information Technology – Document Description and Processing Languages – Office Open XML File Formats – Part 2: Open Packaging Conventions
- O-PAS Standard, Version 2.1: Part 2 – Security, The Open Group Preliminary Standard (P210-2), May 2021, published by The Open Group; refer to: www.opengroup.org/library/p210
- O-PAS Standard, Version 2.1: Part 6.1 – Information and Exchange Models: Overview and Interfaces, The Open Group Preliminary Standard (P210-6a), May 2021, published by The Open Group; refer to: www.opengroup.org/library/p210
- O-PAS Standard, Version 2.1: Part 6.2 – Information and Exchange Models: Basic Configuration, The Open Group Preliminary Standard (P210-6b), May 2021, published by The Open Group; refer to: www.opengroup.org/library/p210
- Unified Architecture (UA) Specification, Part 5 (OPC 10000-5): Information Model (also IEC 62541-5:2015), published by the OPC Foundation; refer to: http://www.opcfoundation.org/UA/Part5/
- Unified Architecture (UA) Specification, Part 6 (OPC 10000-6): Mappings (also IEC 62541-6:2015), published by the OPC Foundation; refer to: http://www.opcfoundation.org/UA/Part6/
- Unified Architecture (UA) Specification, Part 9 (OPC 10000-9): Alarms and Conditions (also IEC 62541-9:2015), published by the OPC Foundation; refer to: http://www.opcfoundation.org/UA/Part9/
- Unified Architecture (UA) Specification, Part 100 (OPC 10000-100): Device Information Model, published by the OPC Foundation; refer to: https://opcfoundation.org/developer-tools/specifications-unified-architecture/part-100-device-information-model/

1.5 Terminology

The key words “shall”, “shall not”, and “should” are used in the O-PAS Standard as defined below. These definitions are taken from IETF RFC 2119.

Shall	Definition is an absolute requirement of the specification.
Shall not	Definition is an absolute prohibition of the specification.
Should	There may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.

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1.6 Future Directions (Informative)

The models in Part 6.3 are expected to be extended to include additional standard alarm types, additional alarm management functions, alerts, events, and other notification types. The definition of the configuration of security related to alarms and roles will be provided in a future version.

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2 Terms and Definitions

For the purposes of the O-PAS Standard, the following terms and definitions apply. Merriam-Webster's Collegiate Dictionary should be referenced for terms not defined in this chapter. The OPAF Glossary and Abbreviations should be referenced for additional terms not defined in this chapter.

The mapping of ANSI/ISA-18.2 terms to OPC UA terms is described in OPC UA Part 9, Table E-1.

2.1 OPAF Glossary and Abbreviations

For the OPAF Glossary and Abbreviations, see the definitions at: www.opengroup.org/open-process-automation-forum/glossary.

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3 Information Model

3.1 Simplified Data Flow Information Model

Figure 1 illustrates the general alarm flow. The O-PAS alarm system is defined by the limits of the red dashed box and the green/shaded boxes highlight the parts of the alarm system to which the configuration descriptions in this document apply.

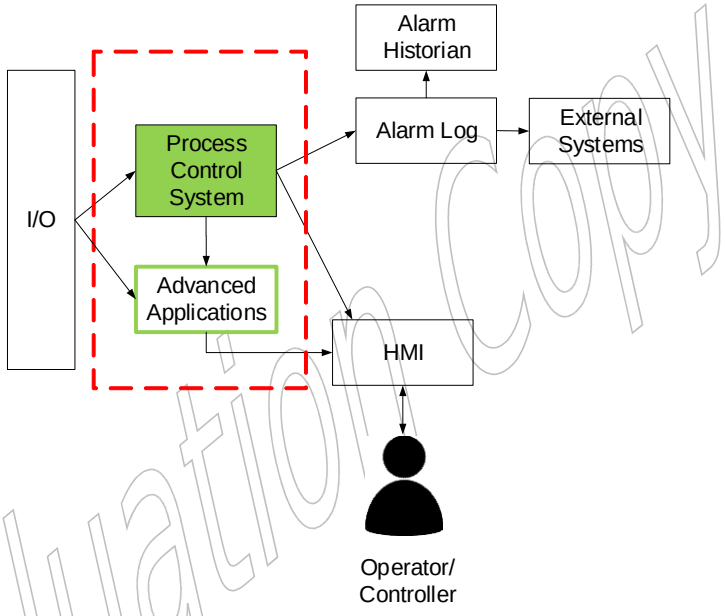

Figure 1: General Alarm Data Flow Model

3.2 Alarm System and Related State Models

OPC UA Part 9 defines implementation of the alarm information model including the alarm state model. The scope of this document is information required for configuration of alarms or the alarm system to support the required functionality.

Configuration data location can be consolidated or distributed. Engineering configuration tools will provide an interface for configuration. At runtime, a system may have multiple alarm engines, which might have individual configuration tools.

In order to configure a system that meets requirements, there are three types of configuration settings to make:

- Alarm attributes common to all alarm types (Section 3.3.2)
- Alarm attributes for specific alarm types (Section 3.3.3)

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- Items related to general alarm execution engine behavior (Section 6.1)

Each of these will be discussed in the relevant sections listed above.

3.3 Alarm Configuration

3.3.1 Overview

Alarm configuration is accomplished through definition of specific alarm attributes. These attributes can reside in function blocks or configuration data structures, but they are not required to do so. This allows for separation of the alarm and logic configuration which enables independent development.

3.3.2 Alarm Attributes Common to all Alarm Types

This includes the definition of the "alarm" object as well as the relation of the alarm object to other functional objects in the O-PAS system. It also includes defining the hierarchy of alarms that the system might generate. Figure 2 provides a graphical view of the common configuration fields in all alarm definitions; for details on the alarm fields see the OPC UA Part 9 Alarm and Condition Model and the OPC UA Part 5 Information Model specification. The green boxes are configured as part of Version 2.1 of the O-PAS Standard. The gray boxes provide optional functionality that can be addressed by a vendor-specific configuration or by a future version of the O-PAS Standard.

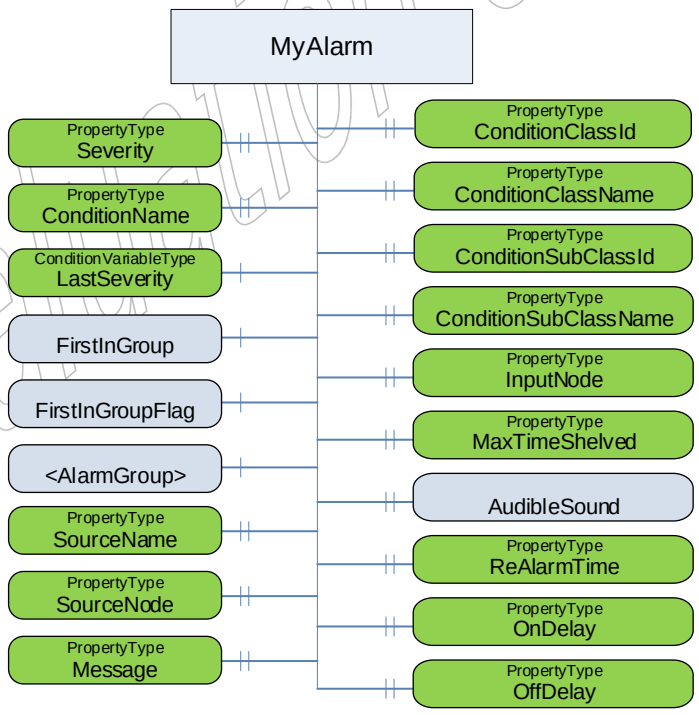

Figure 2: Alarm Instance Configuration

In addition to what is listed in Figure 2, there are a number of optional features that can be included in an alarm (see OPC UA Part 9 for additional details). These optional features do not

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require configuration other than enabling the optional functionality and are not illustrated in the figure.

In addition to the assignment of alarms to a source, the hierarchical notification of alarms can also be configured. Figure 3 illustrates this hierarchical network of alarms. For additional information refer to OPC UA Part 9, Annex B. Figure 3 illustrates that alarms can be obtained by a client from “Area 1” or the client can obtain alarms just related to the “Tank Farm”. The client could also just get alarms from “Device B”.

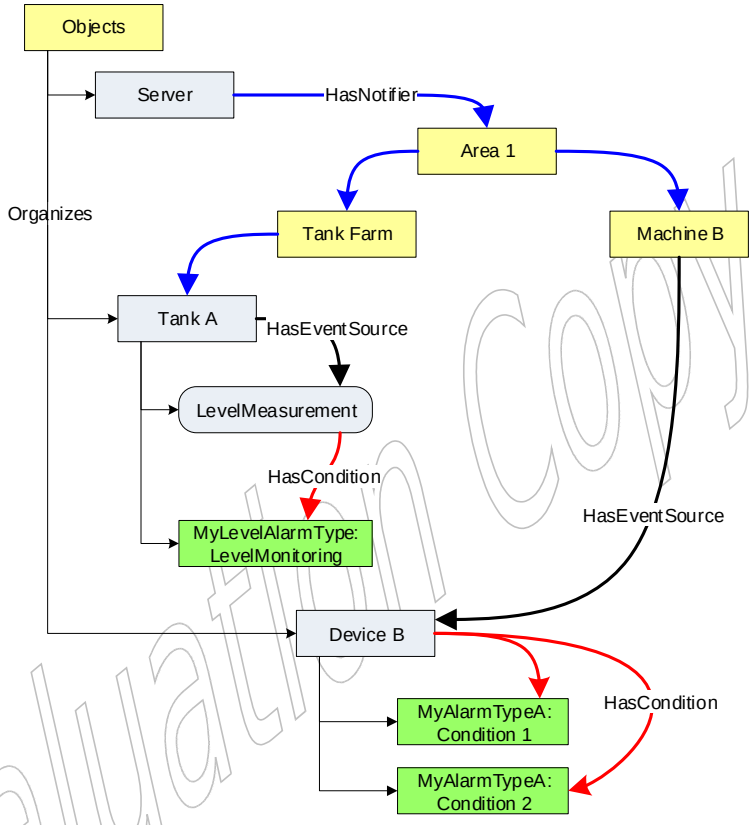

Figure 3: Hierarchical Alarm Reporting

3.3.3 Alarm Attributes for Specific Alarm Types

Specific types of alarms include additional fields for use in configuration. The details of these fields can be found in Appendix A, and are illustrated in Figure 4. The light green boxes illustrate functionality that is optional.

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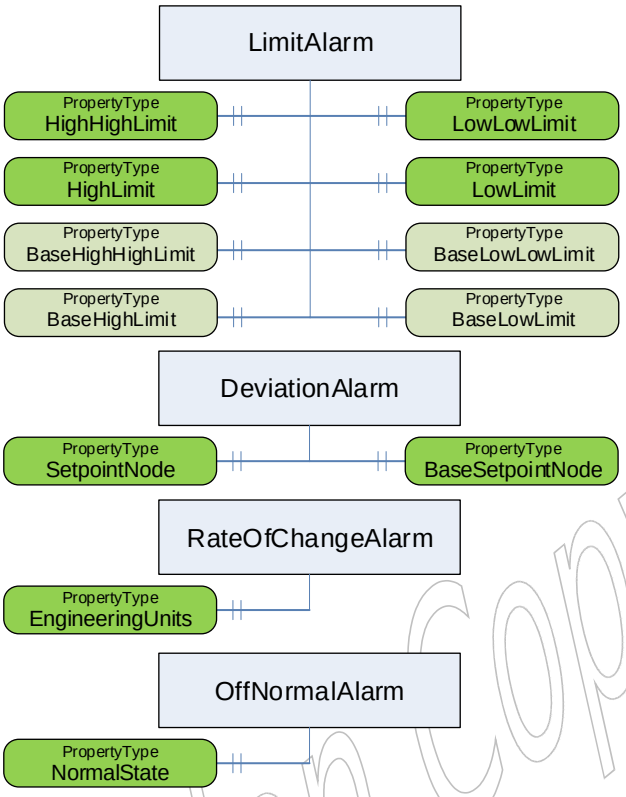

Figure 4: Attribute Fields for Special Alarm Types

For additional detail on these alarm types, see the OPC UA Part 9.

3.3.4 Certification Expiration Alarm Configuration

OPC-UA servers have the ability to provide a message regarding the upcoming expiration of a security certificate. This is specifically the CertificateExpirationAlarmType. Expiration of the certificate could result in loss of data, so this alarm shall be configured and set with an advance notice of no less than two weeks [CRQ-P6.3-0001].

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4 Import/Export Exchange Format

4.1 Overview

The import/export format allows tools to exchange the configuration of alarm-related information used in an O-PAS system. The import/export format is an Open Packaging Conventions (ISO/IEC 29500-2) file (.opas) centered around a .aml file based on the Automation Markup Language (AutomationML) standard (IEC 62714-1), also referred to as AML; that is, an XML-based format that we are using for the exchange of configuration information. For details on the package and AML, see O-PAS Part 6.1.

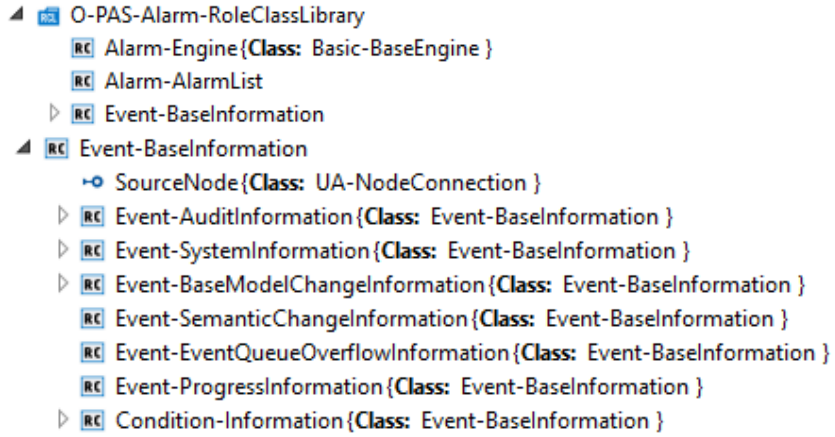
4.2 AML Specifications

4.2.1 Overview

In general, the AML configuration for alarms is closely coupled with the OPC UA model. There are AML role classes for the Alarm-Engine, based on the Basic-BaseEngine, one for the Alarm-AlarmList and one for the Event-BaseInformation and each standard sub-type.

These roles utilize AML attributes and interfaces to reflect the configuration of an alarm in OPC UA.

Figure 5 shows the structure and details of the AML.



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- Event-AuditInformation {**Class:** Event-BaseInformation }
  - Event-AuditSecurityInformation {**Class:** Event-AuditInformation }
    - Event-AuditChannelInformation {**Class:** Event-AuditSecurityInformation }
      - Event-AuditOpenSecureChannelInformation {**Class:** Event-AuditChannelInformation }
    - Event-AuditSessionInformation {**Class:** Event-AuditSecurityInformation }
      - Event-AuditCreateSessionInformation {**Class:** Event-AuditSessionInformation }
        - Event-AuditUrlMismatchInformation {**Class:** Event-AuditCreateSessionInformation }
      - Event-AuditActivateSessionInformation {**Class:** Event-AuditSessionInformation }
      - Event-AuditCancelInformation {**Class:** Event-AuditSessionInformation }
    - Event-AuditCertificateInformation {**Class:** Event-AuditSecurityInformation }
      - Event-AuditCertificateDataMismatchInformation {**Class:** Event-AuditCertificateInformation }
      - Event-AuditCertificateExpiredInformation {**Class:** Event-AuditCertificateInformation }
      - Event-AuditCertificateInvalidInformation {**Class:** Event-AuditCertificateInformation }
      - Event-AuditCertificateUntrustedInformation {**Class:** Event-AuditCertificateInformation }
      - Event-AuditCertificateRevokedInformation {**Class:** Event-AuditCertificateInformation }
      - Event-AuditCertificateMismatchInformation {**Class:** Event-AuditCertificateInformation }
  - Event-AuditNodeManagementInformation {**Class:** Event-AuditInformation }
    - Event-AuditAddNodesInformation {**Class:** Event-AuditNodeManagementInformation }
    - Event-AuditDeleteNodesInformation {**Class:** Event-AuditNodeManagementInformation }
    - Event-AuditAddReferencesInformation {**Class:** Event-AuditNodeManagementInformation }
    - Event-AuditDeleteReferencesInformation {**Class:** Event-AuditNodeManagementInformation }
  - Event-AuditUpdateInformation {**Class:** Event-AuditInformation }
    - Event-AuditWriteUpdateInformation {**Class:** Event-AuditUpdateInformation }
    - Event-AuditHistoryUpdateInformation {**Class:** Event-AuditUpdateInformation }
  - Event-AuditUpdateMethodInformation {**Class:** Event-AuditInformation }
    - Event-AuditConditionInformation {**Class:** Event-AuditUpdateMethodInformation }
      - Event-AuditConditionEnableInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionCommentInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionRespondInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionAcknowledgeInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionConfirmInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionShelvingInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionSuppressionInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionSilenceInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionResetInformation {**Class:** Event-AuditConditionInformation }
      - Event-AuditConditionOutOfServiceInformation {**Class:** Event-AuditConditionInformation }
- Event-SystemInformation {**Class:** Event-BaseInformation }
  - Event-DeviceFailureInformation {**Class:** Event-SystemInformation }
  - Event-SystemStatusChangeInformation {**Class:** Event-SystemInformation }
  - Event-RefreshStartInformation {**Class:** Event-SystemInformation }
  - Event-RefreshEndInformation {**Class:** Event-SystemInformation }
  - Event-RefreshRequiredInformation {**Class:** Event-SystemInformation }
- Event-BaseModelChangeInformation {**Class:** Event-BaseInformation }
  - Event-GeneralModelChangeInformation {**Class:** Event-BaseModelChangeInformation }
- Condition-Information {**Class:** Event-BaseInformation }
  - Condition-DialogInformation {**Class:** Condition-Information }
  - Condition-AcknowledgeableInformation {**Class:** Condition-Information }

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Figure 5: Alarm AML Role Class Overview

Additionally, a series of System Unit Classes (SUCs) are defined for alarms so that they can be instantiated in the InstanceHierarchy; see Figure 6. Note the use of SourceNode, ConditionClassId, ConditionSubClassId, InputNode, and AudibleSound.

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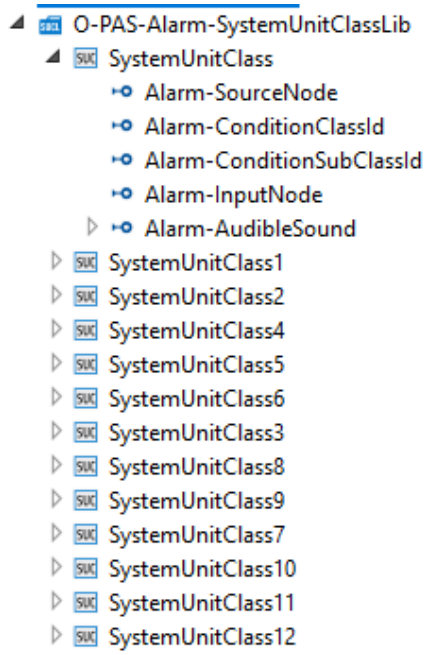

Figure 6: Alarm AML SUC Overview

For formal AML definitions, refer to Appendix B.

4.2.2 AML Interfaces Used for Alarm Configuration

For configuring the OPC UA variables and objects for alarms at runtime, defined interface classes are used. The representation of the various UA links in the AML is as follows:

- Source node: represented using an UA-NodeConnection interface named “SourceNode” with direction “In”

  An internal link from this interface links to the AML element that reflects the OPC UA node to be used as the source node for a BaseEventType object element that reflects the OPC UA object type.

- Input node: represented using an UA-VariableConnection interface named “InputNode” with direction “In”

  An internal link from this interface links to the AML element that reflects the OPC UA variable to be used as the input node for a ConditionType object.

- Setpoint node: represented using an UA-VariableConnection interface named “SetpointNode” with direction “In”

  An internal link from this interface links to the AML element that reflects the OPC UA variable to be used as the setpoint node for an ExclusiveDeviationAlarmType object.

- Base setpoint node: represented using an UA-VariableConnection interface named “BaseSetpointNode” with direction “In”

  An internal link from this interface links to the AML element that reflects the OPC UA variable to be used as the base setpoint node for an ExclusiveDeviationAlarmType object.

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- Normal state: represented using an UA-VariableConnection interface named "NormalState" with direction "In"

  An internal link from this interface links to the AML element that reflects the OPC UA variable to be used as the normal state for an OffNormalAlarmType object.

- Target value node: represented using an UA-VariableConnection interface named "TargetValueNode" with direction "In"

  An internal link from this interface links to the AML element that reflects the OPC UA variable to be used as the target value node for a DiscreptancyAlarmType object.

4.2.3 AML Attributes for Condition Classes

The UPC UA defines a set of standard condition classes that can be used as the ConditionClass/ConditionSubClass(es) for a condition. Additionally, custom condition classes that are sub-types of these standard condition classes can be also defined. Condition classes are used primarily to sort or filter alarms.

To reflect this in an AML file, condition classes are represented as attributes. An attribute type is defined for each of the standard condition classes. Custom condition classes can be defined by creating attribute types that inherit from one of the attribute types defined for the standard condition classes. Alarm instances in the InstanceHierarchy have attributes for type Alarm-ConditionClass or Alarm-SubConditionClass. The Alarm-ConditionClass would have a sub-attribute that is the assigned condition class. The Alarm-SubConditionClass would have one or more sub-attributes that are the assigned condition class(es). Figure 7 shows an example attribute section of an alarm in the InstanceHierarchy. The condition class for this alarm is SystemConditionClassType and the sub-condition classes are TrainingConditionClassType and MaintenanceConditionClassType.

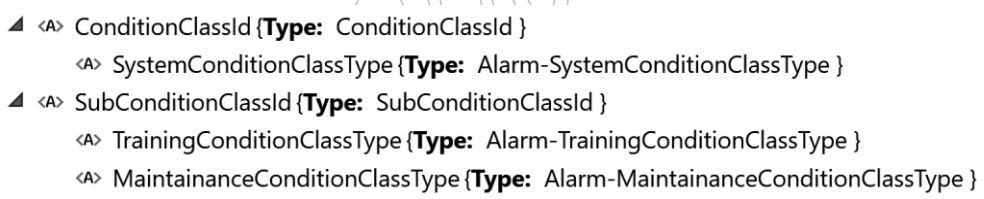

Figure 7: Example of Assigning Condition Classes to an Alarm through AML Attributes

The attribute types reflecting the standard OPC UA condition classes are shown in Figure 8.

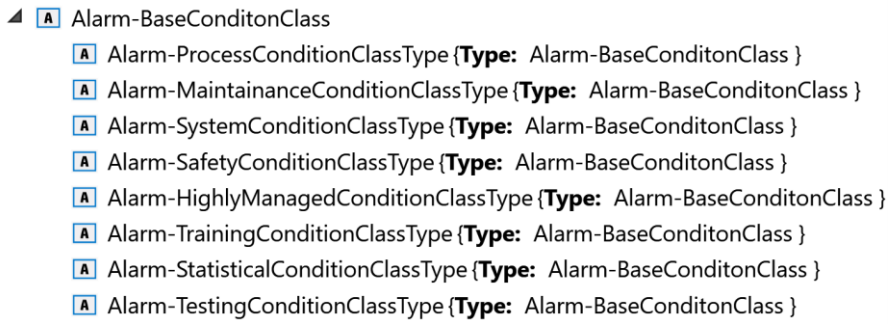

Figure 8: AML Attribute Types for Standard OPC UA Condition Classes

All attributes except the DisplayName and the InstanceHierarchy of the AML file are ignored.

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To represent ConditionClass DisplayName and ConditionSubClass DisplayName for custom condition classes, LocalizedAttribute sub-attributes shall be used for the DisplayName on the instances of the attribute types derived from Alarm-BaseConditionClass [CRQ-P6.3-0002].

4.2.4 Alarms in the AML InstanceHierarchy

This section describes where alarms are placed in the InstanceHierarchy of an AML file and how they are linked to the execution engine responsible for executing them.

4.2.4.1 Alarm-AlarmList Role Class

The Alarm-AlarmList role class is used to provide the alarm list for a project. It groups all the elements with the Event-BaseInformation role (or children of it) together to provide an easy overview of all the alarms used in the project (for configuration purposes in the AML file). This list defines what alarms exist. They then need to be assigned to an alarm engine.

For attributes and the formal definition, see Appendix B.

4.2.4.2 Alarm-Engine Role Class

The Alarm-Engine role class inherits from Basic-BaseEngine and shall be used for execution engines that provide alarms and conditions capability. It shall contain mirrors with the role requirement Event-BaseInformation (or a role inheriting from it). These mirrors reflect the alarms that shall be provided in OPC UA by this execution engine [CRQ-P6.3-0003].

For attributes and the formal definition, see Appendix B.

4.2.4.3 Deployment of Alarms

Alarms can be defined in the alarm list of the AML file, but they still need to be assigned to an execution engine that has the capabilities to provide the alarm functionality at runtime. This "deployment" is done using AML mirrors. Figure 9 shows an example in which the alarm "FI101_Hi" is defined (and configured) in the AML list and then deployed to AlarmEngine_1 in DCN_1. This means that at runtime the alarm FI101_Hi can be found in the OPC UA server of AlarmEngine_1.

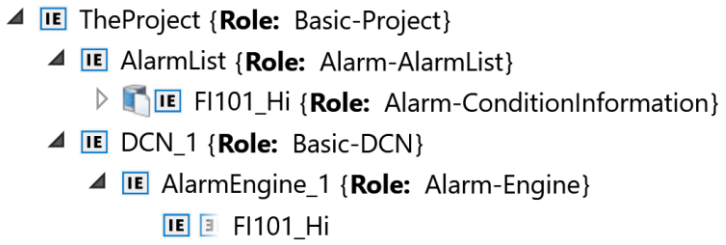

Figure 9: Example of Alarm Deployment in the AML

Note: It is possible that an SUC – e.g., for a function block – comes with preconfigured alarms. The alarm definition would then be part of the instances of this SUC, not the AlarmList. These alarms still need to be mirrored (deployed) into an Alarm-Engine to be available at runtime. (NB: The function block engine might also be an Alarm-Engine.)

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4.3 Profiles and Conformance Units

This section defines the corresponding profiles and conformance units for the OPC UA Information Model for alarms or the AML. Profiles are named groupings of conformance units. Facets are profiles that are combined with other profiles to define the complete functionality of an OPC UA server or client.

No new conformance units are defined in this document since the OPC UA alarm information model is used as-is, with no extensions or changes.

Named "ALM" profiles have dependencies on requirements in other parts of the O-PAS Standard. Those parts contain requirements to implement specific profiles. To be specific, an O-PAS conformant alarm shall conform to the security requirements in the Security Facet (SEC-F-001) described in O-PAS Part 2 – Security [CRQ-P6.3-0012].

Named "ALM" configuration profiles have dependencies on requirements in other parts of the O-PAS Standard. Those parts contain requirements to implement specific profiles. To be specific, an O-PAS conformant alarm shall conform to the import/export profile (OBC-201) described in O-PAS Part 6.2 [CRQ-P6.3-0013].

Named "ALM" execution profiles have dependencies on requirements in other parts of the O-PAS Standard. Those parts contain requirements to implement specific profiles. To be specific, an O-PAS conformant alarm shall conform to the OCF requirements for OCF-001 and optional support for OCF-002. The OCF-F-004 Facet shall be supported [CRQ-P6.3-0014].

O-PAS profiles related to alarm configuration are summarized in Table 1.

Table 1: O-PAS Profiles and Facets

Profile	Related Category	Reference	Conformance Requirement
ALM-001: O-PAS Alarm Server Configuration – Exclusive Alarms	OPAS	Implements configuration for the facets provided in Table 2 and Table 3.	CRQ-P6.3-0001 CRQ-P6.3-0002 CRQ-P6.3-0003 CRQ-P6.3-0012 CRQ-P6.3-0013
ALM-002: O-PAS Alarm Server Configuration – Non-exclusive Alarms	OPAS	Implements configuration for the facets provided in Table 2 and Table 5.	CRQ-P6.3-0001 CRQ-P6.3-0002 CRQ-P6.3-0003 CRQ-P6.3-0012 CRQ-P6.3-0013
ALM-003: O-PAS Alarm Server Execution – Exclusive Alarms	OPAS	Implements execution for the facets provided in Table 2 and Table 3.	CRQ-P6.3-0001 CRQ-P6.3-0010 CRQ-P6.3-0011 CRQ-P6.3-0012 CRQ-P6.3-0014

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Profile	Related Category	Reference	Conformance Requirement
ALM-004: O-PAS Alarm Server Execution – Non-exclusive Alarms	OPAS	Implements execution for the facets provided in Table 3.	CRQ-P6.3-0001 CRQ-P6.3-0010 CRQ-P6.3-0011 CRQ-P6.3-0012 CRQ-P6.3-0014

Table 2, Table 3, and Table 5 provide a list of the available alarm server facets. A server can support multiple facets but is not required to do so.

For a detailed description of these facets, see the OPC Foundation Profiling Tool (available via the links in Table 2, Table 3, and Table 5).

The Conformance Requirement column indicates configuration requirements for the information model described by the OPC UA information related to the provided profile.

Table 2: Alarm Server Facet Common Definition

Conformance Requirement	Profile (Link Provided)	Type	Requirement Status
CRQ-P6.3-0004	A & C Refresh2 Server Facet	Server	M
CRQ-P6.3-0005	A & C Address Space Instance Server Facet	Server	O
CRQ-P6.3-0007	A & C Previous Instances Server Facet	Server	O
CRQ-P6.3-0008	A & C Certificate Expiration Server Facet	Server	M

Table 3: Alarm Server Facet Exclusive Definition

Conformance Requirement	Profile (Link Provided)	Type	Requirement Status
CRQ-P6.3-0006	A & C Refresh2 Server Facet	Server	M

Table 4: Alarm Server Facet Non-Exclusive Definition

Conformance Requirement	Profile (Link Provided)	Type	Requirement Status
CRQ-P6.3-0009	A & C Non-Exclusive Alarming Server Facet	Server	M

Key to Table 2, Table 3, and Table 5

M All “servers” shall implement this facet.

O If “servers” implement this facet, they shall conform to the definition of the facet.

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All functionality designated as “M” shall be implemented as described to be O-PAS conformant. If a product chooses to implement functionality indicated as “O” it shall be implemented to conform to the chosen optional profile, facet, or conformance unit indicated functionality.

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5 Download/Upload Exchange Models

For Version 2.1 of the O-PAS Standard, the download/upload exchange file formats are defined by the vendors' configuration editing or configuration management tools to match their Distributed Control Node (DCN) software.

For Version 2.1 conformant components, vendor tools would use the import/export format for the O-PAS basic configuration defined in Chapter 4 to generate their vendor-specific download/upload format files.

The download files are contained in the AML package file and may be downloaded using the methods defined in O-PAS Part 6.1 – Information and Exchange Models: Overview and Interfaces.

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6 OCF Information Model

6.1 Overview

The O-PAS Standard uses the alarms and conditions information model defined in OPC UA Part 9. It also defines an OPC UA object type for an alarm engine. The OPC UA object type for an alarm execution engine is AlarmEngineType.

Instances of AlarmEngineType shall be the execution engine within a DCN that shall process alarms [CRQ-P6.3-0010]. The formal definition is shown in Table 5.

Table 5: AlarmEngineType

Attribute	**Value**				
BrowseName	AlarmEngineType				
IsAbstract	FALSE				
References	**NodeClass**	**BrowseName**	**DataType**	**TypeDefinition**	**Modeling Rule**
Sub-type of OPASBaseEngineType defined in O-PAS Part 6.2 – Information and Exchange Models: Basic Configuration.					

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7 Namespaces

7.1 Namespace Metadata

Table 6 defines the namespace metadata for this document. The object is used to provide version information for the namespace and an indication about static nodes. Static nodes are identical for all attributes in all servers, including the value attribute. See OPA UA Part 5 for more details.

The information is provided as an object of type NamespaceMetadataType. This object is a component of the namespaces object that is part of the server object. The NamespaceMetadataType object type and its properties are defined in OPC UA Part 5.

The version information is also provided as part of the ModelTableEntry in the UANodeSet XML file. The UANodeSet XML schema is defined in OPC UA Part 6.

Table 6: NamespaceMetadataType Object for this Document

Attribute	**Value**		
BrowseName	https://www.opengroup.org/open-process-automation/O-PAS/Alarm		
References	**BrowseName**	**DataType**	**Value**
HasProperty	NamespaceUri	String	https://www.opengroup.org/open-process-automation/O-PAS/Alarm
HasProperty	NamespaceVersion	String	2.1
HasProperty	NamespacePublicationDate	DateTime	2021-05-17
HasProperty	IsNamespaceSubset	Boolean	True
HasProperty	StaticNodeIdTypes	IdType []	Integer
HasProperty	StaticNumericNodeIdRange	NumericRange []	0,7000
HasProperty	StaticStringNodeIdPattern	String	Null

7.2 Handling of OPC UA Namespaces

Namespaces are used by the OPC UA to create unique identifiers across different naming authorities. The NodeId and BrowseName attributes are identifiers. A node in the UA AddressSpace is unambiguously identified using a NodeId. Unlike NodeIds, the BrowseName cannot be used to unambiguously identify a node. Different nodes may have the same

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BrowseName. They are used to build a BrowsePath between two nodes or to define a standard property.

Servers may often choose to use the same namespace for the NodeId and the BrowseName. However, if they want to provide a standard property, its BrowseName shall have the namespace of the standards body although the namespace of the NodeId reflects something else; for example, the EngineeringUnits property.

Table 7 provides a list of mandatory and optional namespaces used in an O-PAS OPC UA server [CRQ-P6.3-0011].

Table 7: Namespaces used in an O-PAS Alarm Server

NamespaceURI	Description	Use
http://opcfoundation.org/UA/	Namespace for NodeIds and BrowseNames defined in the OPC UA. This namespace shall have namespace index 0.	Mandatory
Local Server URI	Namespace for nodes defined in the local server. This namespace shall have namespace index 1.	Mandatory
http://opcfoundation.org/UA/DI/	Namespace for NodeIds and BrowseNames defined in OPC UA Part 100. The namespace index is server-specific.	Mandatory
http://opcfoundation.org/UA/Dictionary/IRDI	Namespace for NodeIds of International Registration Data Identifiers (IRDIs) using HasDictionaryEntry. The namespace index is server-specific.	Optional
https://www.opengroup.org/open-process-automation/O-PAS/Base	Namespace for NodeIds and BrowseNames defined in O-PAS Part 6.2 – Information and Exchange Models: Basic Configuration. The namespace index is server-specific.	Mandatory
https://www.opengroup.org/open-process-automation/O-PAS/Alarm	Namespace for NodeIds and BrowseNames defined in this document. The namespace index is server-specific.	Mandatory
Vendor specific types	A server may provide vendor-specific types like types derived from object types defined in this document in a vendor-specific namespace.	Optional
Vendor specific instances	A server provides vendor-specific instances of the standard types or vendor-specific instances of vendor-specific types in a vendor-specific namespace. It is recommended to separate vendor-specific types and vendor-specific instances into two or more namespaces.	Mandatory

Table 8 provides a list of namespaces and their index used for BrowseNames in this document. The default namespace of this document is not listed since all BrowseNames without a prefix use this default namespace.

Table 8: Namespaces Used in this Document

NamespaceURI	Namespace Index	Example
http://opcfoundation.org/UA/	0	0:EngineeringUnits
http://opcfoundation.org/UA/DI/	2	2:DeviceRevision

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NamespaceURI	Namespace Index	Example
https://www.opengroup.org/open-process-automation/O-PAS/Base	3	3:ExecutionEngine

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A Alarm Configuration Details and Approach (Informative)

This appendix provides the basic OPC UA-defined alarm types.

A.1 Alarm Types

Table 9 shows the alarm types presented by the FOUNDATION™ Fieldbus (FF) blocks.

Table 9: Typical Alarm Types Based on FF Usage

Ref. Doc				Alarm Type									
	Block Type	Description	Parameter	Standard Block Alarm	High High	High	Dev High	Dev Low	Low	Low Low	State Alarm	Fail	Accept	Ignore
FF-891 4.1	AI	Analog Input		X										
			OUT		X	X			X	X				
FF-891 4.2	DI	Discrete Input		X										
			OUT_D								X			
FF-891 4.3	ML	Manual Loader		X										
			PV		X	X			X	X				
FF-891 4.4	BG	Bias/Gain		X										
FF-891 4.5	CS	Control Selector		X										
FF-891 4.6	PD	Proportional/Derivative		X										
			PV		X	X			X	X				
			SP-PV				X	X						
FF-891 4.7	PID	Prop/Int/Driv		X										
			PV		X	X			X	X				
			SP-PV				X	X						
FF-891 4.8	RA	Ratio		X										
			PV		X	X			X	X				
FF-891 4.9	AO	Analog Output		X										
FF-891 4.10	DO	Discrete Output		X										
FF-892 3.1	DC	Device Control		X								X	X	X
FF-892 3.2	OS	Output Splitter		X										
FF-892 3.3	SC	Signal Characterizer		X										
FF-892 3.4	LL	Lead Lag		X										
FF-892 3.5	DT	Deadtime		X										
FF-892 3.6	IT	Integrator (Totalizer)		X										
FF-892 3.7	SPG	Setpoint Ramp Generator (BKCAL-PRE_OUT)		X			X	X						
FF-892 3.8	ISEL	Input Selector		X										
FF-892 3.9	AR	Arithmetic		X										
FF-892 3.10	TMR	Timer		X										
FF-892 3.11	AAL	Analog Alarm		X										
			OUT		X	X			X	X				
FF-892 3.12	TOT	Totalizer		X										
			OUT			X			X					
FF-893 2	MDI	Multiple Discrete Input		X										
FF-893 3	MDO	Mulitple Discrete Output		X										
FF-893 4	MAI	Mulitple Analog Input		X										
FF-893 5	MAO	Mulitple Analog Output		X										
FF-894 3.1	FOD	Fixed OD		Defined by the DD file										
FF-894 3.2	FPR	Fixed Pgrm Resource		Defined by the DD file										
FF-894 3.3	VOD	Variable OD		Defined by the DD file										
FF-894 3.4	VRB	Variable Pgrm Resource		Defined by the DD file										

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A.2 Mapping FF Alarm Types to OPC UA Alarm Types

Table 10: Mapping of Typical Process Alarms to OPC UA Alarm Types

FF Alarm Type	OPC UA Alarm Type	Comments
Standard Block Alarm	Discrete Alarms (OPC UA Part 9 §5.8.17)	All FF blocks reference the “standard block alarm”. However, that term is not explicitly defined. It can be inferred that the “standard block alarm” relates to the setting of the BLOCK_ERR parameter found in every block, though this relationship is not explicitly stated. Bits in BLOCK_ERR are set based on the many conditions related to the block’s operation that can occur. They are not set by the values processed by the block. FieldComm Group FF-890 §4.4.3.6 defines which bits are valid for each type of function block. We have two options: • Set up a discrete alarm on each bit individually • Define one discrete alarm for the BLOCK_ERR parameter The former gives the most information to the operator. The latter minimizes messages.
High High	Level Alarm (OPC UA Part 9 §5.8.14)	
High	Level Alarm (OPC UA Part 9 §5.8.14)	
Dev High	Deviation Alarm (OPC UA Part 9 §5.8.15)	Based on: • PV above SP (PD, PID) • BKCAL_IN above PRE_OUT (Setpoint Ramp Generator)
Dev Low	Deviation Alarm (OPC UA Part 9 §5.8.15)	Based on: • PV below SP (PD, PID) • BKCAL_IN below PRE_OUT (Setpoint Ramp Generator)
Low	Level Alarm (OPC UA Part 9 §5.8.14)	
Low Low	Level Alarm (OPC UA Part 9 §5.8.14)	
State Alarm	Discrete Alarms (OPC UA Part 9 §5.8.17)	

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FF Alarm Type	OPC UA Alarm Type	Comments
Fail	Discrete Alarms (OPC UA Part 9 §5.8.17)	
Accept	Discrete Alarms (OPC UA Part 9 §5.8.17)	
Ignore	Discrete Alarms (OPC UA Part 9 §5.8.17)	

A.3 Standard Alarm Fields

The following alarm parameters have been identified.

Table 11: Mapping of Standard Alarm Fields to those Provided by the OPC UA

Features of All FF Alarms	OPC UA Approach
Limit	Standard feature of the LimitAlarmType.
Indicator	ActiveState indicator is inherited from AlarmConditionType. All of the proposed OPC UA alarms derive from AlarmConditionType.
Priority/Severity	Standard feature of all alarm types derive from ConditionType. All of the proposed OPC UA alarms derive from ConditionType.
Acknowledgement	Standard feature of all alarm types derived from AcknowledgeableConditionType; e.g., LimitAlarmType, DeviationAlarmType, DiscreteAlarmType.
Alarm Summary	This appears to be a gap in the OPC UA model. A case has been opened with the OPC Foundation to address this gap. See Case #4792.
Alarm Hysteresis	Inherited from NonExclusiveDeviationAlarmType and ExclusiveDeviationAlarmType. All of the proposed OPC UA alarms derive from one of those two types.

A.4 Identifying the Configurable Parameters for the OPC UA Alarms

The O-PAS Standard, Version 2.1 alarm configuration addresses the following items from Figure 12 of OPC UA Part 9:

- Limit alarms and the associated sub-types
- Discrete alarms and the associated sub-types
- Discrepancy alarms and the associated sub-types

Explicitly excluded are SystemEventType messages and the associated sub-types.

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The StateMachineType is not a configured entity so it is not addressed either.

See Figure 10 for an overview of this alarm type derivation.

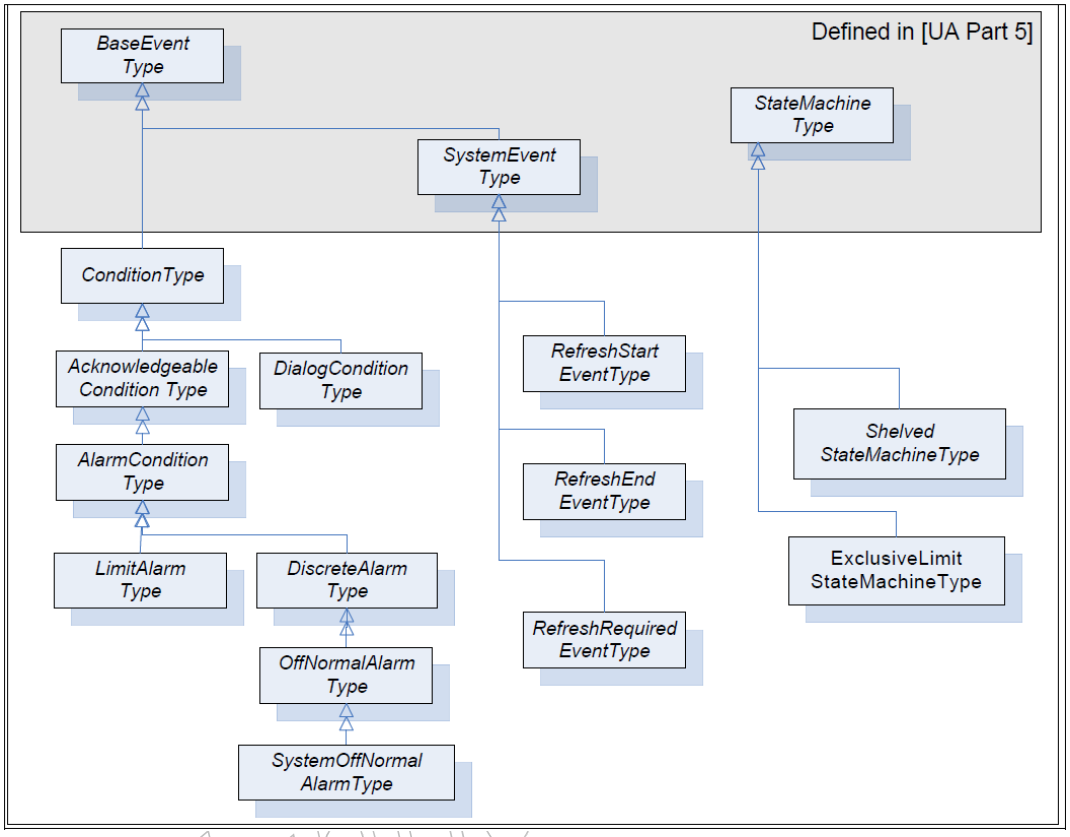

Figure 10: OPC UA Alarm Type Derivation (Overview)

Figure 12 of OPC UA Part 9 (shown below as Figure 11) provides a more complete view. However, it is missing several sub-classes of SystemOffNormalAlarmType.

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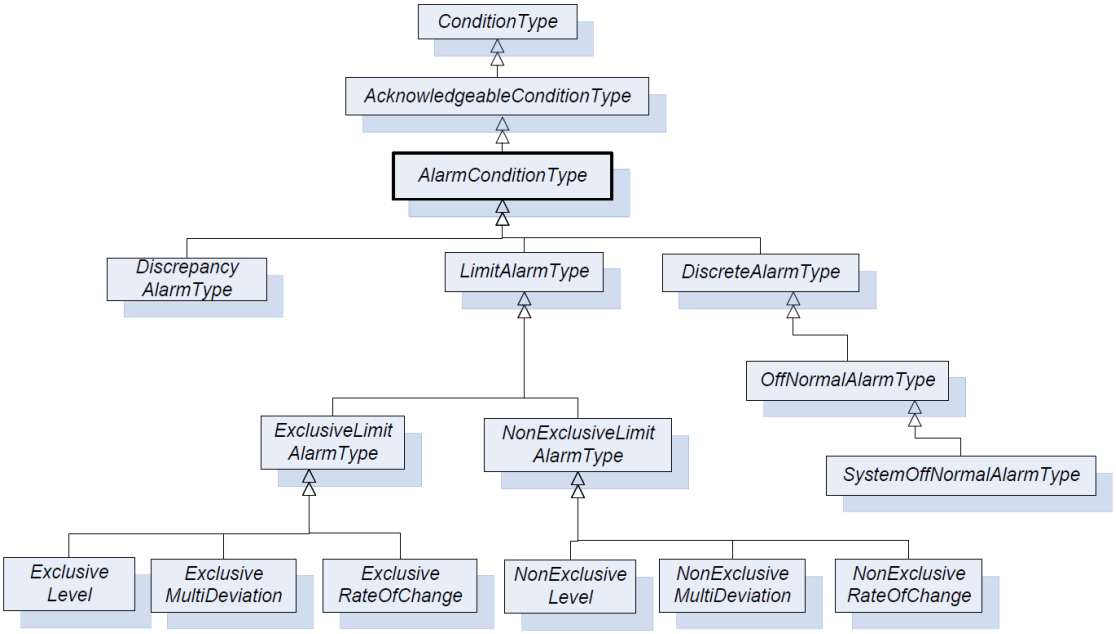

Figure 11: Expanded Derivation of OPC UA Alarm Types

The following tables list the configurable parameters of the various alarm types defined by the OPC Foundation. The first table consists of the configurable items expected for the addressed alarms types. In the table, rows in **Bold** are required. Rows in plain text are alarm parameters that the OPC UA alarm server maintains and supports. Table 12 shows all parameters, configured or not, that appear in all alarm types.

Table 12: Parameters Required for all Process Alarms

UA Type Inheritance	UA References	UA NodeClass	UA BrowseName	UA Data Type	AML Type	UA TypeDefinition	UA Modeling Rule	Source of Value/ Configured
BaseEventType								
	HasProperty	Variable	EventId	ByteString		PropertyType	Mandatory	Server
	HasProperty	Variable	EventType	NodeId		PropertyType	Mandatory	Server
	HasProperty	**Variable**	**SourceNode**	**NodeId**	**AML Interface: UA-NodeConnection**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**SourceName**	**String**	**xs:string**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	Variable	Time	UtcTime		PropertyType	Mandatory	Server
	HasProperty	Variable	ReceiveTime	UtcTime		PropertyType	Mandatory	Server
	HasProperty	Variable	LocalTime	TimeZoneDataType		PropertyType	Optional	Server
	HasProperty	**Variable**	**Message**	**LocalizedText**	**xs:string**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**Severity**	**UInt16**	**xs:unsignedint**	**PropertyType**	**Mandatory**	**Configured**

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UA Type Inheritance	UA References	UA NodeClass	UA BrowseName	UA Data Type	AML Type	UA TypeDefinition	UA Modeling Rule	Source of Value/ Configured
ConditionType								
	HasProperty	**Variable**	**ConditionClassId**	**NodeId**	**Attribute: ConditionClassId with sub-attribute of type Alarm-BaseCondition Class**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	Variable	ConditionClassName	LocalizedText		PropertyType	Mandatory	Server
	HasProperty	***Variable***	***ConditionSubClassId***	***NodeId[]***	**Attribute: SubConditionClassId with sub-attribute(s) of type Alarm-BaseCondition Class**	***PropertyType***	***Optional***	***Configured***
	HasProperty	*Variable*	*ConditionSubClassName*	*LocalizedText[]*		*PropertyType*	*Optional*	*Server*
	HasProperty	**Variable**	**ConditionName**	**String**	**xs:string**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	Variable	BranchId	NodeId		PropertyType	Mandatory	Server
	HasProperty	Variable	Retain	Boolean		PropertyType	Mandatory	Server
	HasComponent	Variable	EnabledState	LocalizedText		TwoStateVariableType	Mandatory	Server
	HasComponent	Variable	Quality	StatusCode		ConditionVariableType	Mandatory	Server
	HasComponent	Variable	LastSeverity	UInt16		ConditionVariableType	Mandatory	Server
	HasComponent	Variable	Comment	LocalizedText		ConditionVariableType	Mandatory	Server
	HasProperty	Variable	ClientUserId	String		PropertyType	Mandatory	Server
AcknowledgeableConditionType								
	HasComponent	Variable	AckedState	LocalizedText		TwoStateVariableType	Mandatory	Server
	HasComponent	***Variable***	***ConfirmedState***	***LocalizedText***	**xs:boolean**	***TwoStateVariableType***	***Optional***	***Feature Configurated***
AlarmConditionType								
	HasComponent	Variable	ActiveState	LocalizedText		TwoStateVariableType	Mandatory	Server
	HasProperty	**Variable**	**InputNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Mandatory**	**Configured**

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UA Type Inheritance	UA References	UA NodeClass	UA BrowseName	UA Data Type	AML Type	UA TypeDefinition	UA Modeling Rule	Source of Value/ Configured
	HasComponent	***Variable***	***SuppressedState***	***LocalizedText***	**xs:boolean**	***TwoStateVariableType***	***Optional***	***Feature Configurated***
	HasComponent	***Variable***	***OutOfServiceState***	***LocalizedText***	**xs:boolean**	***TwoStateVariableType***	***Optional***	***Feature Configurated***
	HasComponent	***Object***	***ShelvingState***		**xs:boolean**	***ShelvedStateMachineType***	***Optional***	***Feature Configurated***
	HasProperty	Variable	SuppressedOrShelved	Boolean		PropertyType	Mandatory	Server
	HasProperty	**Variable**	**MaxTimeShelved**	**Duration**	**xs:duration**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**AudibleEnabled**	**Boolean**	**xs:boolean**	**PropertyType**	**Optional**	***Feature Configurated***
	HasProperty	**Variable**	**AudibleSound**	**AudioDataType**	**xs:string**	**AudioVariableType**	**Optional**	***Feature Configurated***
	HasComponent	**Variable**	**SilenceState**	**LocalizedText**	**xs:string**	**TwoStateVariableType**	**Optional**	**Configured**
	HasProperty	**Variable**	**OnDelay**	**Duration**	**xs:duration**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**OffDelay**	**Duration**	**xs:duration**	**PropertyType**	**Optional**	**Configured**
	HasComponent	Variable	FirstInGroupFlag	Boolean		BaseDataVariableType	Optional	Deferred
	HasComponent	Object	FirstInGroup	AlarmGroupType			Optional	Deferred
	HasAlarmSuppressionGroup	Object	<AlarmGroup>	AlarmGroupType		OptionalPlaceholder		Deferred
	HasComponent	**Object**	**LatchedState**	**LocalizedText**	**xs:boolean**	**TwoStateVariableType**	**Optional**	***Feature Configurated***
	HasProperty	**Variable**	**ReAlarmTime**	**Duration**	**xs:duration**	**PropertyType**	**Optional**	**Configured**
	HasComponent	**Variable**	**ReAlarmRepeatCount**	**Int16**	**xs:unsignedInt**	**BaseDataVariableType**	**Optional**	**Configured**

Notes

1. The Mandatory Modeling Rule means that the OPC UA server requires that the indicated feature be configured for all alarms and conditions.
2. The Optional Modeling Rule provides the user with control over inclusion of optional features for the alarm or condition instance being defined. In this case, the NodeSet excludes the property or component unless the user of the configuration tool specifies that they want the feature. If they want it, they may be required to add additional information; e.g., a value.
3. ConditionClassId and ConditionSubClassId can be assigned to one of the well-known ConditionClassIds; alternatively, additional ConditionClassIds may be generated as part of the configuration and then assigned to alarms.

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Table 13 adds the configurable items required to implement any of the standard “limit alarms”; e.g., high and low, deviation, and rate of change. They are “added” to the parameters shown in Table 12.

Table 13: Parameters Required for Analog (Limit-Based) Process Alarms

Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
LimitAlarmType								
	HasProperty	**Variable**	**HighHighLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**HighLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**LowLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**LowLowLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**BaseHighHigh Limit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**BaseHighLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**BaseLowLimit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**
	HasProperty	**Variable**	**BaseLowLowLi mit**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**

OPA UC Part 9 supports two types of limit alarms:

- Exclusive
- Non-exclusive

Exclusive alarms show only one alarm of a type on the object; e.g., High alarm or High-High alarm, but not both. Non-exclusive alarms show both High and High-High when the occasion requires it. To make this work, an ExclusiveAlarmType is defined from which the three (limit, deviation, rate of change) exclusive alarms are derived. Non-exclusive alarms derive directly from LimitAlarmType and the NonExclusiveLimitAlarmType adds the status bits that are needed. These alarm types are covered in Table 14 and Table 15.

Table 14: Exclusive Alarm Types – Configurable Parameters

Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
ExclusiveLimitAlarmType								
	HasComponent	Object	LimitState			*ExclusiveLimitStateMachineType*	Mandatory	Server
ExclusiveLevelAlarmType								
ExclusiveDeviationAlarmType								

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Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
	HasProperty	**Variable**	**SetpointNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**BaseSetpointNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Optional**	**Configured**
ExclusiveRateOfChangeAlarmType								
	HasProperty	**Variable**	**EngineeringUnits**	**EUInformation**	**AML Interface: UA-UOMConnection**	**PropertyType**	**Optional**	**Configured**

The non-exclusive alarm configurable parameters are defined in Table 15.

Table 15: Non-Exclusive Alarm Types – Configurable Parameters

Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
NonExclusiveLimitlAlarmType								
	HasComponent	Variable	HighHighState	LocalizedText	**xs:Boolean**	TwoStateVariableType	Optional	***Feature Configurated***
	HasComponent	Variable	HighState	LocalizedText	**xs:Boolean**	TwoStateVariableType	Optional	***Feature Configurated***
	HasComponent	Variable	LowState	LocalizedText	**xs:Boolean**	TwoStateVariableType	Optional	***Feature Configurated***
	HasComponent	Variable	LowLowState	LocalizedText	**xs:Boolean**	TwoStateVariableType	Optional	***Feature Configurated***
NonExclusiveLevelAlarmType								
NonExclusiveDeviationAlarmType								
	HasProperty	**Variable**	**SetpointNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**BaseSetpointNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Optional**	**Configured**
NonExclusiveRateOfChangeAlarmType								

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Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
	HasProperty	**Variable**	**Engineering Units**	**EUInformation**	**AML Interface: UA-UOMConnection**	**PropertyType**	**Optional**	**Configured**

Table 16 defines the discrete alarm types. The discrete alarms are all derived from the "OffNormalAlarmType" and only the "CertificateExpirationAlarmType" adds to it. Most of the information related to the certificate alarms comes from the service; only the advanced notice is alarmed.

Table 16: Configurable Parameters for Discrete Alarm Types

Type Inheritance	References	NodeClass	BrowseName	Data Type	XML/AML	TypeDefinition	Modeling Rule	Source of Value/ Configured
DiscreteAlarmType								
OffNormalAlarmType								
	HasProperty	**Variable**	**NormalState**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Mandatory**	**Configured**
SystemOffNormalType								
TripAlarmType								
InstrumentDiagnosticAlarmType								
SystemDiagnosticAlarmType								
CertificateExpirationAlarmType								
	HasProperty	Variable	ExpirationDate	DateTime		PropertyType	Mandatory	
	HasProperty	**Variable**	**ExpirationLimit**	**Duration**	**xs:duration**	**PropertyType**	**Optional**	**Configured**
	HasProperty	Variable	CertificateType	NodeId		PropertyType	Mandatory	
	HasProperty	Variable	Certificate	ByteString		PropertyType	Mandatory	

The last alarm type to be supported is the discrepancy alarm type which is captured in Table 17.

Table 17: Configurable Parameters for Discrepancy Alarms

Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
DiscrepancyAlarmType								

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Type Inheritance	References	NodeClass	BrowseName	Data Type	AML Type	TypeDefinition	Modeling Rule	Source of Value/ Configured
	HasProperty	**Variable**	**TargetValueNode**	**NodeId**	**AML Interface: UA-VariableConnection**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**ExpectedTime**	**Duration**	**xs:duration**	**PropertyType**	**Mandatory**	**Configured**
	HasProperty	**Variable**	**Tolerance**	**Double**	**xs:double**	**PropertyType**	**Optional**	**Configured**

A.4.1 Use of AML RefSemantics

The RefSemantics field in the AML has been used to signal to the tool creating the instructions to the OPC server concerning certain attributes. The strings used and their purpose are defined in Table 18.

Table 18: RefSemantic Setting Details

RefSemantic Setting	Meaning
OPC UA GUID	Should be in the standard GUID format, 8-4-4-4-12. See OPC UA Part 6.
OPC UA LocalizedText	Follows the standard LocalizedText approached used in the AML.
OPC UA 1of4	Used to signal to the tool that builds the configuration for the OPC UA server that only one of the four attributes in this location should be sent to the OPC UA server.
OPC UA Optional	The alarm message field represented by this AML attribute is optional. If the field in the message represented by this attribute is desired, include the attribute in the download to the OPC UA server. The value of the field is supplied by the OPC UA server. If the field in the messages represented by this attribute is not desired, it should be omitted from the download to the OPC UA server. In some cases, there is a following sentence or two to provide more information to the tool user.

Note: If an attribute is marked as Optional in the OPC UA tables, it should have the RefSemantic of “OPC UA Optional”. Attributes marked Mandatory in the OPC UA tables do not have a special RefSemantic setting. They are assumed to be mandatory in the AML; that is, they need to be configured and downloaded.

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B AML Definitions (Normative)

The normative definitions are in the AML files available here:

http://open-process-automation.projects.opengroup.org/twg/part-6-information-models/part-6.3-alarms-events/docx/v02.01.00prelim/opas_part_6_3.aml

The AML files defined for O-PAS Part 6.3 require that the AML file for O-PAS Parts 6.1 and 6.2 are also loaded which provide elements with the prefixes UA- and Basic-.

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C Abbreviations

For the OPAF Glossary and Abbreviations, see the definitions at: www.opengroup.org/open-process-automation-forum/glossary.

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Index

.opas file 9
alarm configuration 6
alarm flow 5
alarm functions 1
alarm object 6
alarm parameters 25
alarm state model 5
alarm types 23
Alarm-AlarmList 14
Alarm-Engine 14
AlarmEngineType 19
alarms and events 1
AML 9
AML files 34
AML mirror 14
base setpoint node 12
CertificateExpirationAlarmType 8
condition class 13
configuration settings 5
conformance 1
conformance units 15
discrepancy alarm 32
discrete alarms 32
download/upload 18
exclusive alarms 30
facets 15
FOUNDATION Fieldbus 23
hierarchy of alarms 6
import/export 1, 18
import/export format 9
input node 12
IRDI 21
namespace 20
NamespaceMetadataType 20
non-exclusive alarms 30
normal state 13
OBC-201 15
OCF-F-004 15
OPC UA alarm 15
OPC UA Part 9 ..... 1, 19
Open Packaging Conventions 9
profiles 15
RefSemantics 33
SEC-F-001 15
security certificate 8
setpoint node 12
source node 12
SUC 11
target value node 13

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